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Batch: Data Analytics Dec Live Batch

Assignment 29: Tableau Basics

Dataset Used Sample – Superstore (Inbuilt dataset in Tableau)

Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis Ans: Tableau:

Tableau is a powerful data visualization and business intelligence tool used to analyze data and create interactive dashboards and reports.

Why Tableau is preferred over traditional spreadsheet tools:

1. It provides **interactive and dynamic visualizations**, unlike static charts in Excel.
2. It can **handle large datasets efficiently**.
3. It has a **drag-and-drop interface**, making it easy to use.
4. It supports **real-time data analysis**.
5. It offers **advanced analytics features** like forecasting and trend analysis.

Industries where Tableau is used:

- Finance
- Healthcare

Use Case:

- In the retail industry, Tableau helps analyze sales data to identify top-performing products and understand customer buying patterns.

Q2. After connecting to the Sample – Superstore dataset:

a) List the main types of fields available (examples: time-based, geographical, numerical).

b) Identify any four Dimensions and any four Measures from the dataset.

Ans:

(a) Types of Fields:

- Time-based fields: Order Date, Ship Date

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- Geographical fields: Country, State, Region
- Numerical fields: Sales, Profit, Quantity
- Categorical fields: Category, Sub-Category **(b) Dimensions and Measures:**

Dimensions:

- Category
- Sub-Category
- Region
- Customer Name **Measures:**
- Sales
- Profit
- Quantity
- Discount

Q3. Create a Bar Chart showing Total Sales by Category.

Explain: Which field goes to Rows Which field goes to Columns

Which aggregation is applied to Sales

Answer: Bar Chart showing Total Sales by Category:

- Columns: Sales
- Rows: Category
- Aggregation: SUM(Sales)

Tableau automatically applies SUM aggregation to the Sales field. Category is used as a dimension to divide data, while Sales is aggregated to show total values.

Q4. What insight can be gained by comparing Sales and Profit together? Then explain how you would create a chart to analyze Profit by Sub-Category.

Answer:

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Comparing Sales and Profit helps identify that high sales do not always mean high profit. Some products may have high sales but low profit due to discounts or costs.

Chart Creation (Profit by Sub-Category):

- Drag Sub-Category to Rows
- Drag Profit to Columns
- Select Bar Chart

This helps identify profitable and loss-making sub-categories.

Q5. Create a Line Chart to show Sales Trend over Order Date. Which date level would you choose (Year / Quarter / Month)? Why are line charts suitable for time-based analysis?

Ans:

- Drag Order Date to Columns
- Change date level to Month
- Drag Sales to Rows **Date Level Chosen:**

Month

Reason:

Monthly data provides a balanced view of trends without too much detail or too little information.

Why Line Charts:

- Show trends over time
- Easy to identify patterns
- Useful for forecasting

Q6. Apply a Basic Filter to display data only for:

Region = “West” Category = “Technology” Explain

the steps and mention which shelf is used.

Answer: steps for Basic Filter to display:

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1. Drag Region to Filters shelf
2. Select West
3. Drag Category to Filters shelf
4. Select Technology **Shelf Used:**

Shelf is used for applying the Filters on Worksheet

This will display data only for the West region and Technology category.

Q7. Explain the business importance of geographical analysis.

Then describe how you would create a Map showing Sales by State using Sample – Superstore Ans:

Business Importance of Geographical Analysis:

Geographical analysis helps organizations understand how their business performs across different locations such as states, cities, or regions.

Importance:

1. **Identify High and Low Performing Areas**
   - Helps businesses know which regions generate more sales or profit.
2. **Better Decision-Making** ○ Companies can focus marketing and resources on weak areas.
3. **Targeted Marketing** ○ Promotions can be customized based on location.
4. **Resource Allocation** ○ Efficient distribution of products and services.
5. **Market Expansion** ○ Helps identify new potential markets

Creating a Map Showing Sales by State in Tableau (Sample – Superstore):

Steps:

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1. Open Tableau and load the **Sample – Superstore dataset**
2. Go to a new worksheet
3. Drag **State → Rows** ○ Tableau will automatically recognize it as a geographical field
4. Drag **Sales → Color (Marks Card)**
5. From the **Marks type**, select **Map** (or Tableau may create it automatically)

Q8. What is the difference between a Worksheet and a Dashboard in Tableau? Why is a dashboard more useful for decision-makers?

Ans:

Worksheet:

A worksheet is a single visualization such as a bar chart or line chart.

Dashboard:

A dashboard is a collection of multiple worksheets displayed together in one view.

Why Dashboard is Useful:

- Shows multiple insights in one place
- Provides better understanding of data
- Interactive and user-friendly
- Helps in faster decision-making

Q9. What is Tableau Prep, and why is it important in real-world data projects? What is Tableau Prep, and why is it important in real-world data projects? Then explain how you would: Remove null values Remove duplicate records Change incorrect data types (Answer in steps, no tool access required.) Ans:

<u>What is Tableau Prep?</u>

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Tableau Prep is a tool used for cleaning, transforming, and preparing data before analysis.

Importance:

- Handles messy real-world data
- Improves data quality
- Ensures accurate analysis
- Saves time **Steps:**

Remove Null Values:

1. Open dataset
2. Identify null fields
3. Apply filter
4. Exclude null values

Remove Duplicate Records:

1. Use Clean Step
2. Select all fields
3. Apply Remove Duplicates

Change Incorrect Data Types:

1. Identify incorrect field
2. Click data type icon
3. Change to correct type (Date, Number, String)

Q10. A retail manager wants answers to the following:

Which category generates the highest profit? Which region performs poorly? How does sales change over time?

a) Mention which charts you would use

b) Identify one key insight that management could derive

c) Explain how this helps in business decision-making Answer:

(a) Charts to Use:

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- Highest Profit by Category → Bar Chart
- Poor Performing Region → Bar Chart / Map
- Sales Over Time → Line Chart **(b) Key Insight:**

Technology category generates highest profit, while some regions perform poorly and need improvement. Sales show variation over time.

(c) Business Decision-Making:

- Focus on profitable categories
- Improve low-performing regions
- Plan inventory based on trends
- Reduce losses and increase efficiency